Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A spring of spring constant $5 \times 10^3 \mathrm{~N/m}$ is stretched initially by $5 \mathrm{~cm}$ from the unstretched position. Then the work required to stretch it further by another $5 \mathrm{~cm}$ is

(A) $12.50 \mathrm{Nm}$

(B) $18.75 \mathrm{Nm}$

(C) $25.00 \mathrm{Nm}$

(D) $6.25 \mathrm{Nm}$

Q.2

When a rubber-band is stretched by a distance x, it exerts restoring force of magnitude $\mathrm{F}=\mathrm{ax}+\mathrm{bx}^2$ where a and b are constants. The work done in stretching the unstretched rubber-band by L is:

(A) $\mathrm{aL}^2+\mathrm{bL}^3$

(B) $\frac{1}{2}\left(aL^2+bL^3\right)$

(C) $\frac{aL^2}{2}+\frac{bL^3}{3}$

(D) $\frac{1}{2}\left(\frac{\mathrm{aL}^2}{2}+\frac{\mathrm{bL}^3}{3}\right)$

Q.3

Two similar springs P and Q have spring constants $\mathrm{K_p}$ and $\mathrm{K_Q}$, such that $\mathrm{K_P}>\mathrm{K_Q}$. They are stretched, first by the same amount (case a,) then by the same force (case b). The work done by the springs $\mathrm{W_p}$ and $\mathrm{W_Q}$ are related as, in case (a) and case (b), respectively

(A) $\mathrm{W_P}=\mathrm{W_Q} ; \mathrm{W_P}=\mathrm{W_Q}$

(B) $\mathrm{W_P}>\mathrm{W_Q} ; \mathrm{W_Q}>\mathrm{W_P}$

(C) $\mathrm{W_P}<\mathrm{W_Q} ; \mathrm{W_Q}<\mathrm{W_P}$

(D) $\mathrm{W_P}=\mathrm{W_Q} ; \mathrm{W_P}>\mathrm{W_Q}$

Q.4

A body of mass $m \mathrm{~kg}$ is taken up by an agent on smooth incline plane of inclination $\theta\left(\sin \theta=\frac{1}{x}\right)$ with constant acceleration $a \mathrm{~m/s^2}$. The final velocity of the body is $\mathrm{vm/s}$. The work done by the agent during this motion is (initial velocity of the body $=0$)

(A) $\frac{1}{2} \mathrm{mv}^2(\mathrm{~g}+\mathrm{xa})$

@IITJEE_Advanced

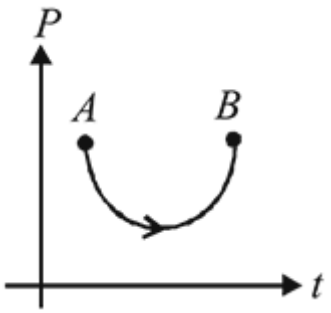
(B) $\frac{mv^2}{a}\left(\frac{g}{2}+a\right)$

(C) $\frac{2mv^2x}{a}(a+gx)$

(D) $\frac{mv^2}{2ax}(g+xa)$

Q.5

The net power of all the forces acting on a particle (P) versus time curve is shown. Work done upon the particle from A to B



(A) increases

(B) decreases

(C) first increases then decreases

(D) first decreases then increases

Q.6

A locomotive of mass m starts moving so that its velocity varies as $v = \alpha s^{2/3}$, where α is constant and s is the distance traversed. The total work done by all the forces acting on the locomotive during the first t second after the start of motion is

(A) $\frac{1}{8}m\alpha^4t^2$

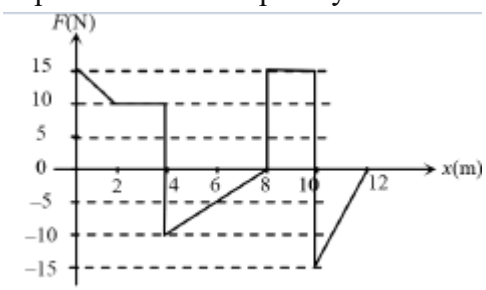
(B) $\frac{m\alpha^6t^4}{162}$

(C) $\frac{m\alpha^6t^4}{81}$

(D) $\frac{m\alpha^4t^2}{2}$

Q.7

A particle is acted upon by a force which varies with position x as shown in the figure. The incorrect statement is



(A) Work done by force F between $x = 0$ to $x = 8$ m is 25 J.

@IITJEE_Advanced

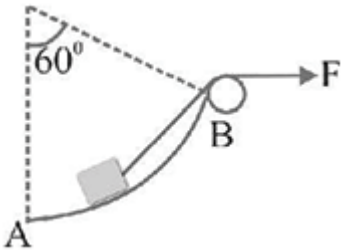
(B) Work dane by force F between $x = 4\ \text{m}$ to $x = 12\ \text{m}$ is $-5\ \text{J}$

(C) Work done by force F is zero at one instant between $x = 0$ to $x = 12\ \text{m}$

(D) Work doneby force F is never zero between $x = 0$ to $x = 12\ \text{m}$

Q.8

A $10\ \text{kg}$ block is pulled in the vertical plane along a frictionless surface in the form of an arc of a circle of radius $10\ \text{m}$. The applied force is of 200 N as shown in figure. If the block had started from rest at A, the velocity at B would be



(A) $1.7\ \text{m/s}$

(B) $17\ \text{m/s}$

(C) $27\ \text{m/s}$

(D) $34\ \text{m/s}$

Q.9

Calculate the work done on the tool by $\vec{F}(11.25\text{i} + 11.25\hat{\text{j}})\text{N}$ if the tool is first moved out along the x -axis to the point $x = 3.00\ \text{m}, y = 0$ and then moved parallel to the y-axis to $x = 3.00\ \text{m}, \text{y} = 3.00\ \text{m}$

(A) $67.5\ \text{J}$

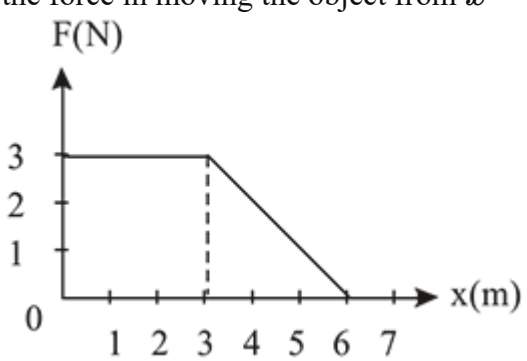
(B) $85\ \text{J}$

(C) $102\ \text{J}$

(D) $7.5\ \text{J}$

Q.10

A force F acting on an object varies with distance as shown here. The force is in N and x in m. The work done by the force in moving the object from $x = 0$ to $x = 6\text{mis}$



(A) $18.0\ \text{J}$

@IITJEE_Advanced

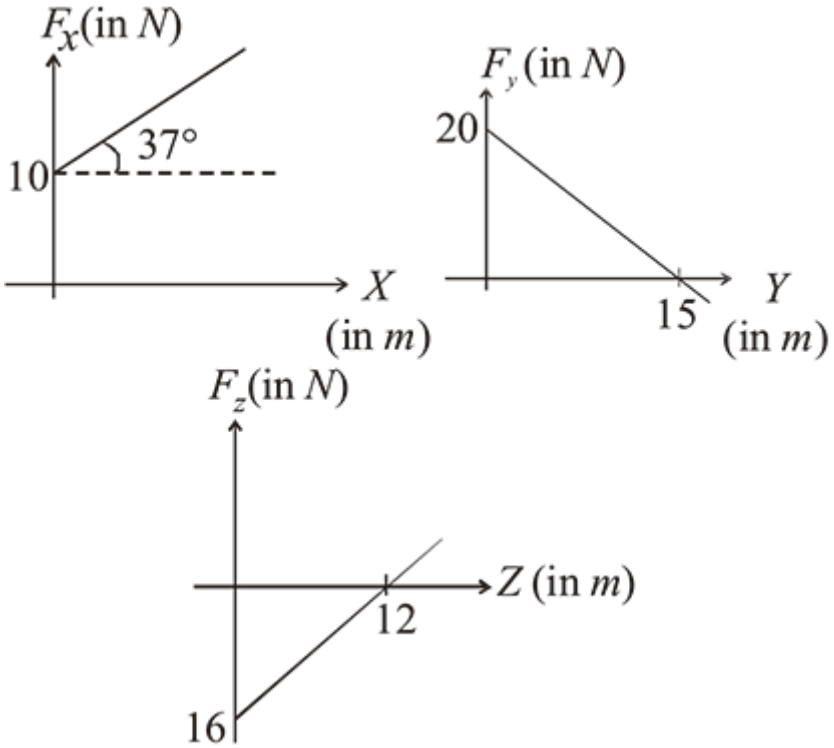
(B) 13.5 J

(C) 9.0 J

(D) 4.5 J

Q.11

The components of a force acting on a particle are varying according to the graphs shown. When the particles move from $(0, 5, 6)$ to $(2, 10, 0)$ then the work done by this force is



(A) 192 J

(B) $\frac{400}{3}$ J

(C) $\frac{287}{2}$ J

(D) None of these

Q.12

A force $F = -K(y\hat{i} + x\hat{j})$ (where K is a positive constant) acts on a particle moving in the xy plane. Starting from the origin, the particle is taken along the positive x axis to the point $(a, 0)$, and then parallel to the y axis to the point (a, a), The total work done by the force F on the particle is

(A) $-2\mathrm{Ka}^2$

(B) $2Ka^2$

(C) $-Ka^2$

(D) Ka^2

Q.13

A cord is used to lower vertically a block of mass M, a distance d at a constant downward acceleration of $g/4$. The

@IITJEE_Advanced

work done by the cord on the block is

(A) $\mathrm{Mg}\frac{\mathrm{d}}{4}$

(B) $3\mathrm{Mg}\frac{\mathrm{d}}{4}$

(C) $-3\mathrm{Mg}\frac{\mathrm{d}}{4}$

(D) Mgd

Q.14

A uniform force of $(3i + j)$ newton acts on a particle of mass $2\ \mathrm{kg}$. The particle is displaced from position $(2\hat{i} + \hat{k})$ meter to position $(4\hat{i} + 3\hat{j} - \hat{k})$ meter. The work done by the force on the particle is

(A) $6\ \mathrm{J}$

(B) $13\ \mathrm{J}$

(C) $15\ \mathrm{J}$

(D) $9\ \mathrm{J}$

Q.15

The position of a particle of mass $4\ \mathrm{g}$, acted upon by a constant force is given by $x = 4t^2 + t$, where x is in metre and t in second. The work done during the first 2 seconds is

(A) $128\ \mathrm{mJ}$

(B) $512\ \mathrm{mJ}$

(C) $576\ \mathrm{mJ}$

(D) $144\ \mathrm{mJ}$

Q.16

A body moves a distance of $10\ \mathrm{m}$ along a straight line under the action of a force of 5 newtons. If the work done is 25 joules, the angle which the force makes with the direction of motion of body is

(A) 0°

(B) 30°

(C) 60°

(D) 90°

Q.17

A force acts on $\mathrm{a30gm}$ particle in such a waythat the position of the particle as a function of time is given by $x = 3t - 4t^2 + t^3$, where x is in metres andtisin seconds. The work done during the first 4 seconds is

(A) $5.28\ \mathrm{J}$

(B) $450\ \mathrm{mJ}$

@IITJEE_Advanced

(C) $490\ \text{mJ}$

(D) $530\ \text{mJ}$

Q.18

The work done by a force $\vec{F} = \left(-6x^3\hat{i}\right)\ N$, in displacing a particle from $x = 4m$ to $x = -2m$ is

(A) $360J$

(B) $240J$

(C) $-240J$

(D) $-360J$

Q.19

Work done by static friction on an object

(A) may be positive

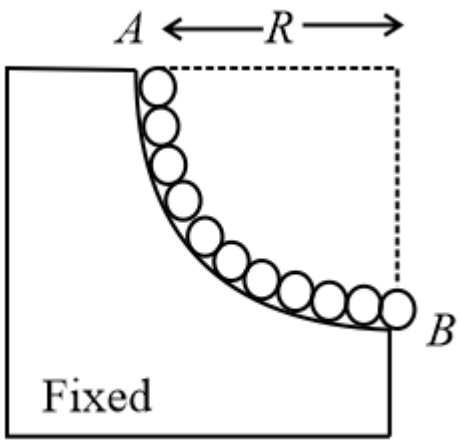
(B) must be negative

(C) must be zero

(D) None of these

Q.20

As shown in the figure, a chain of mass m is placed on a smooth quarter circular portion of radius R. End A is tied with a wedge while the remaining chain is free, then the minimum work required to be done by the external agent to make the chain horizontal keeping point A fixed, is



(A) $mgR\left(1 - \frac{2}{\pi}\right)$

(B) $mg\frac{2\text{R}}{\pi}$

(C) $mg\sqrt{2}\frac{2\text{R}}{\pi}$

(D) None

@IITJEE_Advanced

Answers & Solutions

Q.1 Answer:

$18.75\mathrm{Nm}$

Solution:

$$k = 5 \times 10^3 \ \mathrm{N/m}$$

$$W = \frac{1}{2}k\left(x_2^2 - x_1^2\right) = \frac{1}{2} \times 5 \times 10^3 \left[(0.1)^2 - (0.05)^2\right]$$

$$= \frac{5000}{2} \times 0.15 \times 0.05 = 18.75\mathrm{Nm}$$

Q.2 Answer:

$\frac{aL^2}{2} + \frac{bL^3}{3}$

Solution:

Work done in stretching the rubber-band by a distance dx is

$$dW = Fdx = \left(ax + bx^2\right)dx$$

Integrating both sides,

$$W = \int_0^L axdx + \int_0^L bx^2dx = \frac{aL^2}{2} + \frac{bL^3}{3}$$

Q.3 Answer:

$\mathrm{W_P} > \mathrm{W_Q}; \mathrm{W_Q} > \mathrm{W_P}$

Solution:

As we know work done in stretching spring $\mathrm{w} = \frac{1}{2}\mathrm{kx}^2$

where k **= spring constant** x **= extension Case (a) If extension (x) is same,** $\mathrm{W} = \frac{1}{2}\ \mathrm{Kx}^2$

So, $W_\mathrm{P} > \mathrm{W_Q}$

$(\because \mathrm{K_P} > \mathrm{K_Q})$

Case (b) If spring force (F) is same $\mathrm{W} = \frac{\mathrm{F}^2}{2\,\mathrm{K}}$

$\mathrm{So}, \mathrm{W_Q} > \mathrm{W_P}$

Q.4 Answer:

$\frac{\mathrm{mv}^2}{2\mathrm{ax}}(\mathrm{g} + \mathrm{xa})$

Solution:

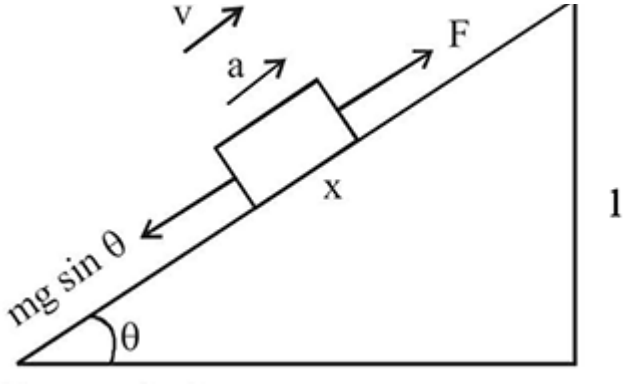
$\sin\theta = \frac{1}{x}$

From free body diagram of the body

$\mathrm{F} - \mathrm{mg}\sin\theta = \mathrm{ma}$

$\mathrm{F} = \mathrm{m}(\mathrm{g}\sin\theta + \mathrm{a}) = \mathrm{m}\left(\frac{\mathrm{g}}{\mathrm{x}} + \mathrm{a}\right)$

@IITJEE_Advanced



Displacement of the body till its velocity reaches v $v^2 = 0 + 2as \Rightarrow s = \frac{v^2}{2a}$
Now, work done $= \mathrm{Fs}\cos 0^\circ = \frac{\mathrm{m}}{\mathrm{x}}(\mathrm{g} + \mathrm{ax}) \times \frac{\mathrm{v}^2}{2\mathrm{a}}$
$= \frac{\mathrm{mv}^2}{2\mathrm{ax}}(\mathrm{g} + \mathrm{ax})$

Q.5 Answer:

increases

Solution:

Area under graph increases. Hence work done upon the particle from A to B increases.

Q.6 Answer:

$\frac{m\alpha^6 t^4}{162}$

Solution:

$i = \alpha s^{2/3}$
$\frac{ds}{dt} = \alpha s^{2/3}$
$\int_D^a \frac{ds}{s^{2/3}} = \int_0^t \alpha dt \Rightarrow 3s^{1/3} = \alpha t$
$s = \frac{\alpha^3}{27}t^3 \Rightarrow ds = \frac{\alpha^3}{9}t^2 dt$
$v = \frac{\alpha^3}{9}t^2$
$W = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{\alpha^3 t^2}{9}\right)^2 = \frac{m\alpha^6 t^4}{162}$

Q.7 Answer:

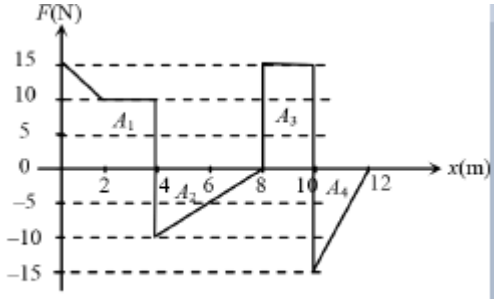
Work done by force F is zero at one instant between
$x = 0$ to $x = 12$ m

Solution:

$A_1 = \frac{1}{2} \times 5 \times 2 + 4 \times 10 = 45$ J
$A_2 = \frac{-1}{2} \times 4 \times 10 = -20$ J
$A_3 = 15 \times 2 = 30$ J
$A_4 = \frac{-1}{2} \times 2 \times 15 = -15$ J
$W(x = 0$ **to** $x = 8$ m$) = A_1 + A_2 = 45 - 20 = 25$ J
$W(x = 4$ m **to** $x = 12$ m$) = A_2 + A_3 + A_4 = -5$ J
As $|A_1| > |A_2|$ **and** $|A_1 + A_3| > |A_2 + A_3|$

@IITJEE_Advanced

$\therefore$ **Work done was never zero between** $x = 0$ **to** $x = 12$ m



Q.8 Answer:

17 m/s

Solution:

$W_{\text{gravity}} + W_F = \Delta K$
$-10 \times g\,(10 - 10\cos 60^\circ) + 200 \times 10 \sin 60^\circ$
$= \frac{1}{2} \times 10 \times v_f^2 - 0 \quad \therefore \quad v_f = 17$ m/s

Q.9 Answer:

67.5 J

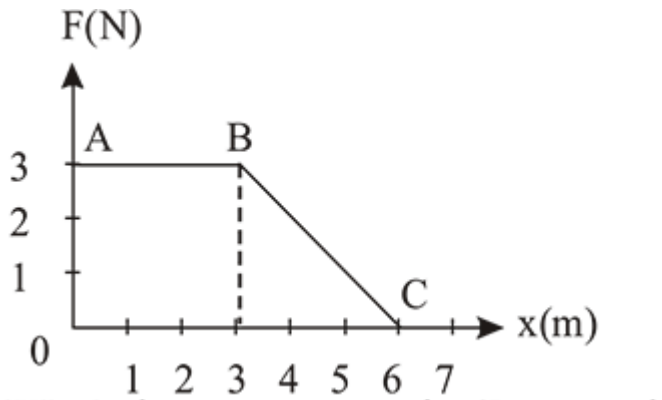
Solution:

Net displacement is $(3\hat{i} + 3\hat{j})$ so work done $= \overrightarrow{F} \cdot \Delta\overrightarrow{s} = 11.25(\hat{i} + \hat{j}) \cdot (3\hat{i} + 3\hat{j})$
$= 33.75 \times 2 = 67.50$ J

Q.10 Answer:

13.5 J

Solution:



Work done = area under F − x **graph = area of trapezium** OABC $= \frac{1}{2}(3 + 6)(3)$
$= 13.5$ J

Q.11 Answer:

$\frac{287}{2}$ J

Solution:

@IITJEE_Advanced

From given graph :

$\overrightarrow{\mathrm{F}}=\left(\frac{3}{4}\mathrm{x}+10\right)\hat{\mathrm{i}}+\left(20-\frac{4}{3}\mathrm{y}\right)\hat{\mathrm{j}}+\left(\frac{4}{3}\mathrm{z}-16\right)\hat{\mathrm{k}}$

$\mathrm{W}=\int\overrightarrow{\mathrm{F}}\cdot\overrightarrow{\mathrm{ds}}$

$=\int_{(0,5,6)}^{(2,10,0)}\left(\frac{3}{4}x+10\right)\hat{i}+\left(20-\frac{4}{3}y\right)\hat{j}+\left(\frac{4}{3}z-16\right)\hat{k}$

$[dx\hat{i}+dy\hat{j}+dz\hat{k}]=\frac{287}{2}\ \mathrm{J}$

Work done can also be found by finding area under these curves.

Q.12 Answer:

$-Ka^2$

Solution:

(c) The expression of work done by the variable force F on the particle is given by

$W=\int\vec{F}\cdot\overrightarrow{d\ell}$

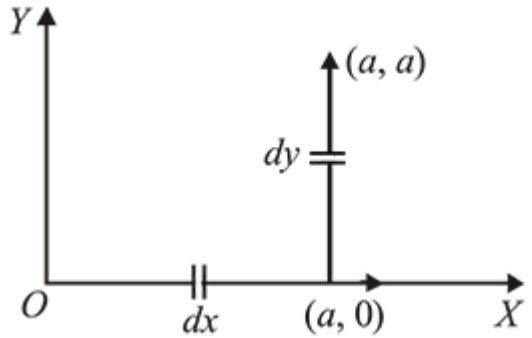
In going from $(0,0)$ to $(a,0)$, the coordinate of varies from 0 to $'a'$, while that of y remains zero. Hence, the work done along this path is :

$W_1=\int_0^a(-Kx\hat{j})\cdot dx\hat{i}=0$

$[\because\hat{j}\cdot\hat{i}=0]$

In going from $(a,0)$ to (a,a) the coordinate of x remains constant $(=a)$ while that of y changes from 0 to a'. Hence, the work done along this path is $W_2=$

$\int_0^a\left[(-K(y\hat{i}+a\hat{j})\cdot dy\hat{j}\right]=ka\int_0^a dy=-Ka^2$



Hence, $W=W_1+W_2=-ka^2$

Q.13 Answer:

$-3\mathrm{Mg}\frac{\mathrm{d}}{4}$

Solution:

As the cord is trying to hold the motion of the block, work done by the cord is negative.

$$\mathrm{W}=-\mathrm{M}(\mathrm{g}-\mathrm{a})\mathrm{d}$$

$$=-\mathrm{M}\left(\mathrm{g}-\frac{\mathrm{g}}{4}\right)\mathrm{d}=\frac{-3\mathrm{Mgd}}{4}$$

@IITJEE_Advanced

Q.14 Answer:

$9\ \mathrm{J}$

Solution:

Given : $\overrightarrow{F} = 3\hat{i} + \hat{j}$

$\overrightarrow{r_1} = (2\hat{i} + \hat{k}), \overrightarrow{r_2} = (4\hat{i} + 3\hat{j} - \overrightarrow{k})$

$\overrightarrow{r} = \overrightarrow{r_2} - \overrightarrow{r_1} = (4\hat{i} + 3\hat{j} - \overrightarrow{k}) - (2\hat{i} + \hat{k})$

or $\overrightarrow{r} = 2\hat{i} + 3\hat{j} - 2\hat{k}$

So work done by the given force $w = \overrightarrow{f} \cdot \overrightarrow{r} = (3\hat{i} + \hat{j}) \cdot (2\hat{i} + 3\hat{j} - 2\hat{k}) = 6 + 3 = 9\ \mathrm{J}$

Q.15 Answer:

$576\ \mathrm{mJ}$

Solution:

here, $m = 4,\ g = 4 \times 10^{-3}\ \mathrm{kg}$

$$x = 4t^2 + t \quad \therefore \frac{dx}{dt} = 8t + 1 \frac{d^2x}{dt^2} = 8$$

Work done, $W = \int f dx = \int m \frac{d^2x}{dt^2} \left(\frac{dx}{dt}\right) dt$

$= \int_0^2 \left(4 \times 10^{-3}\right)(8)(8t + 1) dt$

$= 32 \times 10^{-3} \int_0^2 (8t + 1) dt = 32 \times 10^{-3} \left[\frac{8t^2}{2} + t\right]_0^2$

$= 32 \times 10^{-3} \left[4(2)^2 + 2 - 0\right] = 576\ \mathrm{mJ}$

Q.16 Answer:

60°

Solution:

$W = Fs\cos\theta, \cos\theta = \frac{W}{Fs} = \frac{25}{5 \times 10} = \frac{1}{2},$

$\theta = 60^\circ$

Q.17 Answer:

$5.28\ \mathrm{J}$

Solution:

The mass is $30\ \mathrm{g}$ **or** $0.03\ \mathrm{kg}$ **and the position of the particle is** $x = 3t - 4t^2 + t^3$**. The velocity is given as,**

$\frac{dx}{dt} = 3 - 8t + 3t^2$

The acceleration is given as,

$\frac{d^2x}{dt^2} = -8 + 6t$

The work done is given as,

@IITJEE_Advanced

$$dW = (ma)dx$$
$$dW = (0.03) \times (-8 + 6t)\left(3 - 8t + 3t^2\right) dt$$
$$W = (0.03) \int_0^4 \left(18t^3 - 72t^2 + 82t - 24\right) dt$$
$$W = (0.03) \left(18 \times \frac{t^4}{4} - 72 \times \frac{t^3}{3} + 82 \times \frac{t^2}{2} - 24t\right)_0^4$$
$$W = (0.03) \times 176$$
$$W = 5.28 \text{ J}$$

Thus, the work done during the first 4 s is 5.28 J.

Q.18 Answer:

$360J$

Solution:

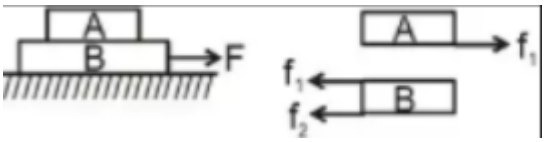
$$W = \int_{x_1}^{x_2} F dx = \int_4^{-2} -6x^3 dx$$
$$= -6 \left| \frac{x^4}{4} \right|_4^{-2} = \frac{-6}{4} \left[(-2)^4 - (4)^4\right] = 360 \text{ J}.$$

Q.19 Answer:

may be positive

Solution:

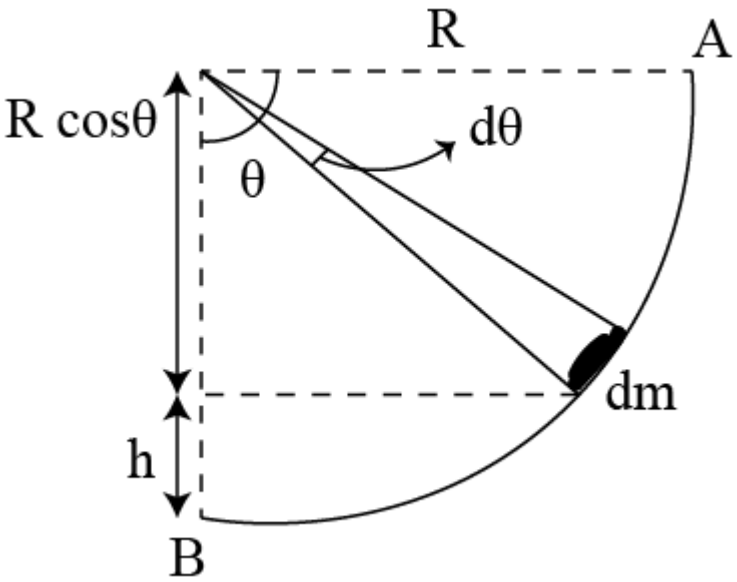
Work done by static friction can be negative, positive as well as zero. Consider the blocks shown in the figure to be moving together due to friction between them. The free body diagrams of both the blocks is shown below. Work done by static friction on A is positive and on B is negative.



Q.20 Answer:

$mgR\left(1 - \frac{2}{\pi}\right)$

Solution:



@IITJEE_Advanced

Let a small element of mass dm subtend angle $d\theta$ at the center.

$\therefore dm = \frac{m}{\frac{\pi}{2}} d\theta = \frac{2m}{\pi} d\theta$

So, the potential energy.

$$dv = dmgh = \frac{2m}{\pi} d\theta \cdot g \cdot (h - R\cos\theta)$$

$$\therefore U_i = \frac{2mgh}{\pi} \int_0^{\pi/2} (1 - \cos\theta) d\theta$$

$$\Rightarrow V_i = \frac{2mgR}{\pi} \left[[\theta]_0^{\frac{\pi}{2}} - [\sin\theta]_0^{\frac{\pi}{2}} \right]$$

$$= \frac{2mgR}{\pi} [\pi/2 - 1]$$

$$U_i = \mathrm{mgR} \left[1 - \frac{2}{\pi} \right];$$

and $U_f = 0$ (after being horizontal)

$\therefore$ So, a minimum of $mgR\left(1 - \frac{2}{x}\right)$ amount of work should be done by an external agent;

@IITJEE_Advanced